

ASCII / hex dump

Logs, ROMs, and memory dumps are often rows of hex with a printable side view

Bytes, glyphs, and controls

- Printable ASCII typically starts at hex twenty, the space
- Newline is line feed, hex zero A
- A C-style string ends with NUL, hex zero zero
- When a dump cannot show a glyph, it prints a dot so the columns still line up
- Hex on the left and glyphs on the right are two views of the same bytes

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

ASCII / hex dump literacy

Read a classic **hex dump**: offsets, byte hex, and printable ASCII — and map characters like 0x0A and 0x20.

Starter example: Buffer starts with Hi, a newline (0x0A), OK, then NUL. Read the dump's offset, hex, and ASCII columns together.

Load starter example

Challenges 0 / 22 cleared

Quiz: printable: Printable ASCII range starts at? Answer: 0x20

Answer

Show hintCheckNextIdle

Quiz: printableQuiz: LFQuiz: NULQuiz: dotStarter HiFind LFFind NULWrite LFWrite spaceWrite NULLoad HelloQuiz: 'A'See ABC0xFF is dotEdit byteQuiz: columnsFill zerosQuiz: CRExplain spaceQuiz: UTF-8Offset 0Full dump

Core ideas

Three columns

Offset · hex bytes · ASCII glyph (or . if not printable).

Printable

Usually 0x20–0x7E. Controls like LF=0x0A show as ..

ASCII hex dump starter

learn_digital — ascii hex

Workbook practice

- In the workbook track, write a tiny dump by hand
- Encode the letters H and i, then a line feed, then O and K, then NUL
- Mark which offsets are printable and which would show as dots
- Name one RTL or bring-up pitfall

Pitfalls to watch

- Do not assume every byte is a letter, controls and high bytes break that story
- Do not confuse hex “zero A” with the digit characters for one and zero
- And remember: the browser lab is literacy
- Real dumps from simulators and flash still use the same byte-to-glyph rules

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, finish a few challenges after the starter
- On paper, dump a short string with one control and one NUL
- When you are ready, take the short quiz, then continue to Gray code